



Question(s): 2/9

Geneva, 06-14 Sep 2022

CONTRIBUTION

Source: Ministry of Communications, India; Telecom Regulatory Authority of India, India; Telecommunications Standards Development Society, India; Centre for Development of Telematics, India

Title: Proposed A.1 justification for proposed draft new ITU-T Recommendation J.FRCAS "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content"

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Abstract: This contribution contains the proposed A.1 justification for a proposed draft new ITU-T Recommendation on the functional requirements of Conditional Access Systems (CAS) deployed in Digital Addressable Systems (DAS) for television broadcasting to enable factual subscriber-base reporting and to ensure protected delivery of content to authorised subscribers.

The objective of the proposed Recommendation is to address two major concerns related to CAS, namely, underreporting of subscribers and content piracy, leading to revenue loss to broadcasters, content providers and the governments.

This draft new Recommendation will elaborate the various functional requirements of the CAS such as log requirements, reports requirements, database requirements, security requirements, service requirements, and more. Compliance to these requirements in CAS performance will address the two concerns mentioned above.

Proposal: The proposed A.1 justification is placed as Annex-1. The proposed baseline text is annexed with another contribution and it provides suggestive clauses for this proposed new recommendation on "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content".

Annex 1

A.1 justification for proposed draft new ITU-T J.FRCAS "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content"

Question:	Q2/9	Proposed new ITU-T Recommendation	Online or Geneva, 06-14 September 2022	
Reference and title:	ITU-T J.FRCAS "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content"			
Base text:	SG9-Cnnn		Timing:	2023-08
Editor(s):	Avinash Agarwal, Ministry of Communications, India avinash.70@gov.in Rakesh S. Desai, Ministry of Communications, India dircb2.tec-dot@gov.in		Approval process:	AAP
Scope (defines the intent or object of the Recommendation and the aspects covered, thereby indicating the limits of its applicability): The proposed Recommendation will specify the functional requirements of Conditional Access Systems (CAS) deployed in Digital Addressable Systems (DAS) for television broadcasting to enable factual subscriber-base reporting and to ensure protected delivery of content to authorised subscribers. The objective of the proposed Recommendation is to address two major concerns related to CAS, namely, underreporting of subscribers and content piracy, leading to revenue loss to broadcasters, content providers and the governments. This draft new Recommendation will elaborate the various functional requirements of the CAS such as log requirements, reports requirements, database requirements, security requirements, service requirements, and more. Compliance to these requirements in CAS performance will address the two concerns mentioned above.				
Summary (provides a brief overview of the purpose and contents of the Recommendation, thus permitting readers to judge its usefulness for their work): Most countries have implemented Digital Addressable Systems (DAS) for distribution platforms such as Cable TV, Direct to Home (DTH), Internet Protocol Television (IPTV), and Headend In The Sky (HITS). DAS enables subscribers to have a choice and option to select their channels and packages. Conditional Access System (CAS) and Subscriber Management System (SMS) are integral parts of a DAS environment. While CAS is responsible for encrypting the content and its delivery in a secure and encrypted manner only to authorised subscribers, SMS handles the subscriber management functions. The quality of service depends on the CAS and SMS systems deployed by the Distribution Platform Owner (DPO). Since the content providers and broadcasters have no control over the CAS and SMS, as these are deployed at the DPO end, they often complain about the loss of revenue due to various malpractices, such as underreporting of subscriber-base, distribution of channels to unauthorised users, and content piracy. Several countries have enabling provisions for mandatory audit of the DPO systems to provide requisite assurance to broadcasters and content providers, and CAS and SMS systems have to conform to certain technical features to check piracy and other malpractices. Multiple CAS solutions are available, some using advanced embedded security while others using sub-standard security solutions. Systems having sub-standard solutions can be prone to manipulations and vulnerable to hacking, leading to loss of revenue to the concerned broadcasters and the government. Therefore, it is necessary to frame benchmarks for the CAS systems, prescribing technical specifications focusing on functional requirements to ensure factual reporting of subscriber-base and protected delivery of content to authorised subscribers. Creating a framework that prevents the deployment of sub-standard CAS will bring preventive control from potential threats and revenue losses arising due to the vulnerability of such systems, bring economic efficiency, improve quality of service, and improve the end-consumer experience.				

Question 2/9 broadly deals with the technical specifications of CAS. Therefore, this contribution proposes a new work item to frame such Recommendations. The proposed baseline text annexed with another contribution provides suggestive clauses for this new recommendation.
Relations to ITU-T Recommendations or to other standards (approved or under development): [ITU-T J.93], [ITU-T J.95] These are very old ITU-T recommendations and their scope does not include the functional requirements for the purpose mentioned above. [ITU-R BT.1852-1] This ITU-R Recommendation covers the technical aspects of scrambling and access control but does not include the functional requirements to prevent malpractices and manipulations mentioned above.
Liaisons with other study groups or with other standards bodies: None
Supporting members that are committing to contributing actively to the work item: Ministry of Communications, India; Telecom Regulatory Authority of India, India; Telecommunications Standards Development Society, India; Centre for Development of Telematics, India